

Software Requirements Specification

ChurnGuard — Churn-Risk Detection & Retention Workflow

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Version	1.0
Standard followed	IEEE 830 structure, adapted
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1. Introduction

1.1 Purpose

This SRS defines the system context, external interfaces, data requirements and non-functional characteristics of ChurnGuard. It complements FRD-CG-001, which holds the functional behaviour, and FSD-CG-001, which holds the build-level design.

1.2 Product scope

ChurnGuard is an extension to the existing Zoho CRM Plus estate. It adds a scheduled scoring service, three custom objects, an account-level risk panel, an automation layer for task routing and SLA tracking, and an analytics layer for portfolio reporting. It introduces no customer-facing interface in Release 1.

1.3 Definitions

Term	Definition
Composite risk score	Weighted 0–100 value expressing churn risk for an account on a given date
Signal	One of five measurable dimensions contributing to the composite score
Sub-score	Normalised 0–100 risk value for a single signal
Tier	Stable / Watch / Critical classification derived from the composite score
Escalation	Movement of an account into a higher-risk tier between cycles
Retention Case	Structured record of an intervention from identification to outcome
Shadow run	Period in which scoring executes and stores results but generates no alerts
High-Value segment	Accounts meeting the ARR threshold agreed with Finance

1.4 References

BRD-CG-001 v1.0 · FRD-CG-001 v1.0 · FSD-CG-001 v1.0 · RTM-CG-001 · UAT-CG-001 · Enterprise CRM Role & Profile Standard · Data Retention Policy

2. Overall description

2.1 System context



ChurnGuard is a consumer of data, not a system of record for it. The CRM Account remains the master record; ChurnGuard writes only risk, task, case and audit data.

2.2 User classes

Class	Volume	Technical skill	Frequency	Primary need
CSM	~15	Low-medium	Daily	Know which accounts need attention today and why

CS Manager	3	Medium	Daily	Team risk position, SLA compliance, escalations
VP Customer Success	1	Low	Weekly	Portfolio position, ARR at risk, save rate
Business Administrator	2	Medium-high	Monthly	Tune weights, thresholds, playbooks safely
Data & BI Analyst	2	High	Ad-hoc	Validate signals, analyse accuracy, tune the model

2.3 Operating environment

Zoho CRM Plus (Accounts, Contacts, Deals, Activities, Desk) · Zoho Analytics for reporting · Deluge for in-platform automation · n8n for scheduled orchestration and cross-system retrieval · SQL data mart for usage and marketplace telemetry · Browser-based access, desktop and tablet; CRM mobile app for task consumption only.

2.4 Design and implementation constraints

Ref	Constraint
C-1	No new platform licence in R1; build within existing Zoho CRM Plus and Analytics entitlements
C-2	Scoring must execute inside the 02:00-04:00 IST batch window
C-3	Platform API call limits apply per day; scoring must use bulk operations rather than per-record calls
C-4	Deluge execution time limits apply per function; scoring must be chunked
C-5	No customer PII is transmitted outside the CRM boundary for scoring
C-6	Role and profile model must extend the existing CRM standard, not bypass it

3. External interface requirements

3.1 Data interfaces

ID	Interface	Direction	Method	Frequency	Payload	Failure handling
IF-01	Usage Data Mart	Inbound	Scheduled query, bulk extract	Daily, pre-scoring	Account ID, active days, orders processed, listings published, modules used — 30d and 90d	Retry ×3 at 15-min intervals; on final failure mark S1 Insufficient Data (FR-007) and raise ops alert
IF-02	Marketplace Connector Health Service	Inbound	REST, paged	Daily	Account ID, connector, state, last sync, error count, attempt count, marketplace health flag	As IF-01, affecting S2
IF-03	Support Desk	Inbound	Platform-native module read	Daily	Account ID, ticket counts by priority, SLA breaches, reopens, latest CSAT	As IF-01, affecting S3
IF-04	Billing / RevOps sync	Inbound	Existing CRM sync, read	Daily	Plan, seats, prior plan/seats, failed payments, oldest overdue invoice age, renewal date	As IF-01, affecting S4
IF-05	CRM core objects	Bidirectional	Native	Continuous	Account, Contact, Activity, Deal, Owner	Native platform handling
IF-06	Analytics workspace	Outbound	Scheduled sync	Post-scoring, daily	Score, sub-scores, tier, movement, case and task data	Dashboard shows last successful refresh timestamp (FR-046)
IF-07	Notification channel	Outbound	Email + in-app + Slack	Event-driven	Alert summary and deep link	Fall back to in-app only; log delivery failure

3.2 User interface requirements

ID	Requirement
UI-01	The account risk panel shall render within the standard Account detail view without a page reload.
UI-02	Risk tier shall be conveyed by label and by shape or icon in addition to colour, so it is distinguishable without colour perception.
UI-03	All screens shall be usable at 1366×768 and above; the task list and risk panel shall remain usable on tablet at 1024px.
UI-04	Numeric values shall show their unit and comparison period inline — for example, "18 active days vs 41 (90-day average)".
UI-05	Any suppressed or reduced-signal score shall carry a visible status label, never a silent omission.
UI-06	Interface text shall follow the existing CRM terminology set; "account", "case" and "task" retain their platform meanings.

4. Data requirements

4.1 Logical data model

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Account (existing)
| 1      1 |      1 |
|          |      |
| n      n |      n |
Risk_Score_History  Retention_Case  Retention_Task
|                  | 1          |
| n                |          |
Signal_Detail      Case_Action_Log
                        SLA_Tracking
Config_Version (standalone, effective-dated)
Audit_Entry    (standalone, immutable)

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4.2 Key entities

Risk_Score_History — one row per account per scoring cycle.

Attribute	Type	Notes
Score_Date	Date	Part of composite key with Account_ID
Account_ID	Lookup	Mandatory
Composite_Score	Integer 0–100	Mandatory
Tier	Picklist	Stable / Watch / Critical
Previous_Tier	Picklist	Null on first cycle
Movement	Picklist	No Change / Escalation / De-escalation
Dominant_Driver	Picklist	One of the five signals
Data_Quality_Flag	Picklist	Complete / Reduced / Suppressed
Config_Version_ID	Lookup	Which configuration produced this score

Signal_Detail — five rows per score record: signal code, sub-score, weight applied, weighted contribution, raw input values (JSON), comparison period, insufficient-data flag.

Retention_Case — account, opening score and tier, dominant driver, playbook applied, owner, status, opened date, closed date, closure outcome, closure reason, linked renewal result.

Retention_Task — account, case link, assignee, created date, due date, SLA tier, first-touch timestamp, elapsed business hours, SLA status.

Config_Version — effective date, weights, tier boundaries, hysteresis, SLA durations, playbook mapping, changed-by, prior values.

4.3 Data quality rules

ID	Rule
DQ-01	Every source record must resolve to exactly one CRM Account ID; unresolved records go to an exception report and are excluded, not guessed.
DQ-02	A feed older than 48 hours is stale; its signal is marked Insufficient Data.
DQ-03	Null is never treated as zero. A missing usage value is missing data, not zero usage.
DQ-04	Composite scores are never back-dated or recalculated retrospectively; a configuration change applies forward only.
DQ-05	Score history is immutable once written.

4.4 Retention and archival

Score history and signal detail: 13 months live, then archived. Retention cases and tasks: retained per CRM record policy. Audit entries: retained 7 years, immutable. Archival is a scheduled job with a reconciliation count logged.

5. Non-functional requirements

5.1 Performance

ID	Requirement	Verification
NFR-01	A full scoring cycle for the in-scope account base shall complete within 90 minutes, inside the 02:00–04:00 IST window.	Timed batch run at production volume
NFR-02	The design shall sustain NFR-01 at three times the current in-scope account volume without architectural change.	Load test at 3× volume
NFR-03	The account risk panel shall render within 3 seconds at the 95th percentile.	Performance test
NFR-04	An on-demand single-account rescore shall complete within 30 seconds.	Performance test
NFR-05	Dashboard views shall return within 8 seconds at the 95th percentile with all filters applied.	Performance test

5.2 Availability and reliability

ID	Requirement
NFR-06	ChurnGuard inherits the CRM platform availability target; no additional single point of failure shall be introduced.
NFR-07	A failed scoring cycle shall be automatically retried once; a second failure shall raise an operations alert within 15 minutes and preserve the prior day's scores.
NFR-08	Scoring shall be idempotent — re-running a cycle for the same date shall produce one history record per account, not duplicates.
NFR-09	Partial failure of one source interface shall not prevent scoring on the remaining signals, subject to FR-008 and FR-009.

5.3 Security and privacy

ID	Requirement
NFR-10	Access shall be governed by CRM roles and profiles; no ChurnGuard object shall be readable by a profile without an explicit grant.
NFR-11	Field-level security shall restrict composite score and signal detail to CS roles and above.
NFR-12	All data in transit between systems shall use TLS 1.2 or higher.
NFR-13	No customer PII shall be included in scoring payloads leaving the CRM boundary; accounts are referenced by internal ID only.
NFR-14	Audit entries shall be immutable and shall not be editable or deletable by any application role, including administrators.
NFR-15	Configuration changes shall require the Business Administrator permission and shall be logged with before and after values.

5.4 Usability and accessibility

ID	Requirement
NFR-16	A CSM shall be able to determine why an account is at risk from the risk panel without navigating to a second screen.
NFR-17	Tier shall never be communicated by colour alone (see UI-02).
NFR-18	All interactive elements shall be keyboard reachable with a visible focus state.
NFR-19	A new CSM shall be able to complete the alert-to-case workflow after a single 30-minute training session; verified in UAT.

5.5 Maintainability and supportability

ID	Requirement
NFR-20	Weights, thresholds, SLAs and playbook mappings shall be configuration, not code.
NFR-21	Scoring logic shall be documented such that a signal can be added or removed without redesigning the scoring engine.
NFR-22	Every scoring cycle shall write a run log: start, end, accounts scored, accounts suppressed, exceptions, config version.
NFR-23	Exception reports shall be retrievable by support without developer involvement.

5.6 Compliance and audit

ID	Requirement
NFR-24	The solution shall comply with the organisation's data retention policy for customer records.
NFR-25	The audit trail shall be sufficient to reconstruct, for any historical date, the score an account held and the configuration that produced it.

6. Verification approach

Requirement class	Method
Data ingestion and scoring	SQL reconciliation of engine output against independently calculated expected values on a 6-month historical sample
Tiering and alerting	Scenario-based functional tests using seeded accounts at each boundary
Dashboard	Figure-by-figure reconciliation against source queries
Performance	Timed runs at production and 3x volume
Security	Role-based access matrix test, one login per profile
Usability	UAT observation with CSMs, no prompting permitted

7. Open items at issue of v1.0

Ref	Item	Owner	Needed by
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O-1	Confirm public holiday calendar source for business-day SLA calculation	CS Ops	Sprint 3
O-2	Confirm archival storage location for 13-month-plus score history	Data Engineering	Sprint 4
O-3	Agree operations alert recipients for scoring cycle failure	CRM Engineering	UAT entry